

Communication Made Possible: A Comprehensive Web Application for Two-Way Sign Language Conversion

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Abstract: Despite technological advances, the struggle with communication barriers for the deaf and of hearing persists. This study introduces a unique Two-Way Language Converter Website (TWSLCW) to help bridge the gap between sign language speakers and those unfamiliar with sign language. By using state-of-the-art machine learning algorithms, the website converts spoken words to sign language and vice-versa instantly. This article discusses the creation, execution, and operation of TWSLCW, highlighting its user-friendly design and accessible features. Additionally, a thorough evaluation assesses the accuracy, effectiveness, and user contentment of the converter. The findings are promising, with participants indicating a high level of satisfaction and success in using TWSLCW for communication. This study contributes to progress in inclusive technology solutions and emphasizes the significance of using technology to improve communication access for various communities.

Keywords: Communication Inclusivity Solution, Interactive Sign Language Platform, Barrier-Free Communication Tool, Empowering Deaf Communication.

I. INTRODUCTION

In a world with more digital communication, the barriers facing people who use sign language remain a big challenge. The deaf and hardly hearing community often encounters a variety of obstacles in communicating effectively with non-signers, affecting their participation in daily life activities, like education, employment, and social interactions. While technology has driven accessibility innovations, creating efficient two-way sign language communication tools persists as an ongoing necessity.

This article deals with this urgent need by exploring and executing a Two-Way Sign Language Converter Website (TWSSCLW). The TWSSCLW presents a very new method of bridging the communication divide between sign language users and non-signers in real time, providing a promising resolution to enhance inclusivity and accessibility in digital communication platforms.

This provides a review of the importance of TWSSCLW in facilitating communication for sign language users. It points out the challenges faced by the deaf and hardly hearing community in old-school communication environments and emphasizes the potential of technology to effectively handle these challenges. Besides, it outlines the goals of the research paper, such as designing, implementing, and assessing the TWSSCLW, to evaluate its efficacy, usability, and impact on communication accessibility.

Through this study, our goal is to contribute to the improvement of inclusive technology solutions that empower sign language users to communicate effectively in various situations. By utilizing modern machine learning algorithms and user-focused design principles, the TWSSCLW offers a very promising path to promoting communication equality and improving the quality of life for the deaf and hardly hearing community.

II. EXISTING SYSTEM

The difference in communication between language users and non-users does not create serious problems that hinder interaction and understanding. Traditional translation methods rely on human agents or pre-written data and often lack accuracy and precision. There is also a lack of effective systems that can instantly and precisely translate spoken or written words into hand gestures. Current solutions also struggle to cope with the nuances and complexities of sign language, leading to errors and misunderstandings. Therefore, there is an urgent need for advanced machine learning models to accurately predict and interpret hand gestures and spoken/written messages. Addressing this challenge could transform access to communication for the deaf and hard-of-hearing community and promote greater inclusion and understanding across language backgrounds.

III. PROPOSED SYSTEM

The proposed system for two-way sign language conversion aims to revolutionize communication accessibility for the deaf and hard-of-hearing community by leveraging advanced technology. This system will employ state-of-the-art machine learning algorithms and computer vision techniques to enable real-time translation of sign language gestures into spoken language and vice versa. The user interface will be intuitively designed to accommodate both sign language users and non-signers, offering visual feedback for sign gestures and providing output through text or synthesized speech. The system's multimodal input and output capabilities will cater to diverse user preferences, ensuring accessibility through customizable features such as font sizes, contrast options, and language settings. Additionally, the system will be scalable and adaptable for deployment across various environments, including educational institutions, healthcare facilities, and public spaces, with the goal of promoting inclusivity and breaking down communication barriers in everyday interactions. Integration with existing technologies will further enhance interoperability and usability, making this system a comprehensive solution for fostering equitable communication experiences for all individuals.

Proposed diagram

GESTURE TO TEXT CONVERSION:

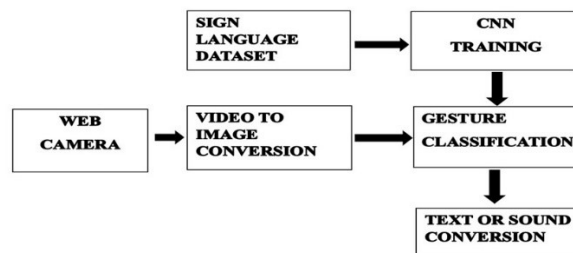


Fig 3.1: Gesture to Text Conversion

VOICE TO GESTURE CONVERSION:

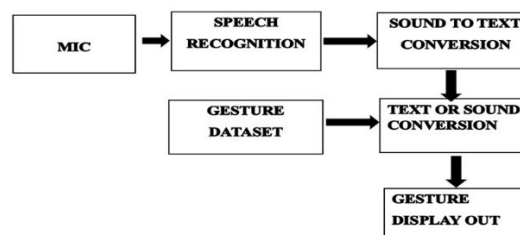


Fig 3.2: Voice to Gesture Conversion

IV. LITERATURE SURVEY

[1]. “Sign Language Recognition System Using Indian Sign Language”–

Deaf and mute people communicate via hand gestures, i.e., sign language, which makes it difficult for nondeaf and non-mute people to understand their language. The problem statement here is to devise a technique for bridging the communication gap between the deaf/dumb community and the general public. The user's hand gestures are detected in real time. An efficient machine learning algorithm will be used to train the model to identify the letters of the ISL and thus converting it to a text format accessible to the normal people. This application will also enable the normal people to convey their message in terms of speech which will further be converted to text so that the latter will be able to receive their message.

[2]. “A Novel Approach of Sign Recognition for Indian Sign Language”

In this approach first gesture is converted into text and then voice format, So normal people understand the meaning of that gesture. Second, Voice is converted into text and then gesture, so physically impaired people can understand it. In this vision based approach used with the dynamic gesture. Pre-processing, Segmentation, Feature extraction, Classification these four are the main steps to recognize hand gesture. In this approach key frame extraction with histogram analysis is used for extract the main frame which has a sign & removes the unwanted frame.

[3]. “Automatic Indian Sign Language Recognition for Continuous Video Sequence”

Sign Language Recognition has become the active area of research nowadays. This paper describes a novel approach towards a system to recognize the different alphabets of Indian Sign Language in video sequence automatically. The proposed system comprises of four major modules: Data Acquisition, Pre-processing, Feature Extraction and Classification. Pre-processing stage involves Skin. Filtering and histogram matching after which Eigen vector based Euclidean distance based Classification Technique was used. 24 different alphabets were considered in this paper where 96% recognition rate was obtained.

[4]. “Recognition of Indian Sign Language using Hand Gestures and Facial Expressions”

This paper presents implementation and analysis of the real time translation of Indian sign language gestures into text offering people with hearing and/or speech impaired, a means of communication with people incapable of understanding sign language. Sign language brings hearing and hearing disabled people together. Also helps communicate with one who has communication disability and the world. The aim of the paper is to develop a Sign Language Recognition (SLR) system that convert Indian Sign Language (ISL) gestures into text. The raw input image which is directly fed from webcam is preprocessed and the face and hands are tracked and segmented using blob analysis, a method for analyzing the movement of the head and hands based on the identification of skin color. Trained data set are created and classification is done by multiclass SVM which predicts the gesture fed in the web cam.

[5]. “Sign Language Recognition for Deaf Sign User”

Sign language recognition is one of the most growing fields of research today and it is the most natural way of communication for the people with hearing problems. A hand gesture recognition system can provide an opportunity for deaf persons to communicate with normal people without the need of an interpreter or intermediate. We are going to build a systems and methods for the automatic recognition of Marathi sign language. Through that we are providing teaching classes for the purpose of training the deaf sign user in Marathi. The system does require hand to be properly aligned to the camera and does not need any special colour markers, glove or wearable sensors. A large set of samples has been used in proposed system to recognize isolated words from the standard Marathi sign language which are taken in front of camera by different deaf sign user. In our proposed system, we intend to recognize some very basic elements of sign language and to translate them to text and vice versa.

V. MODULE DESCRIPTION

Sign Language Gesture Recognition Module: This module will focus on capturing and interpreting sign language gestures using computer vision techniques. It will involve preprocessing the input video stream to isolate hand movements and extract relevant features. Machine learning algorithms, such as convolutional neural networks (CNNs), will be trained on labeled sign language datasets to classify and recognize different gestures accurately.

Text to Sign Language Translation Module: This module will translate text input into sign language gestures. It will utilize natural language processing (NLP) techniques to preprocess and tokenize the text input. The tokenized text will then be fed into a deep learning model, such as a recurrent neural network (RNN) or transformer architecture, trained to generate corresponding sign language gestures.

Audio to Text Conversion Module: This module will convert audio input, such as spoken language, into text. It will involve preprocessing the audio signal, extracting relevant features using techniques like spectrogram analysis, and then utilizing automatic speech recognition (ASR) algorithms to transcribe the audio into text.

Text to Audio Translation Module: This module will convert text input into audio output, facilitating communication with individuals who rely on spoken language. It will involve synthesizing human-like speech from the text using text-to-speech (TTS) synthesis techniques, such as concatenative synthesis or neural network-based approaches.

VI. IMPLEMENTATION AND RESULTS

6.1 Building Architecture

Our work over the past months focused on the design and training of neural networks that effectively use our growing dataset. We studied several architectures to come up with a solution that both meets our high-performance requirements and creates minimal runtime overhead. In the end we converged on an architecture that contains a three-dimensional convolutional network (3D-CNN) to extract spatiotemporal features, a recurrent layer (LSTM) to model longer temporal relations, and a SoftMax layer that outputs class probabilities.

6.2 Preparing Dataset

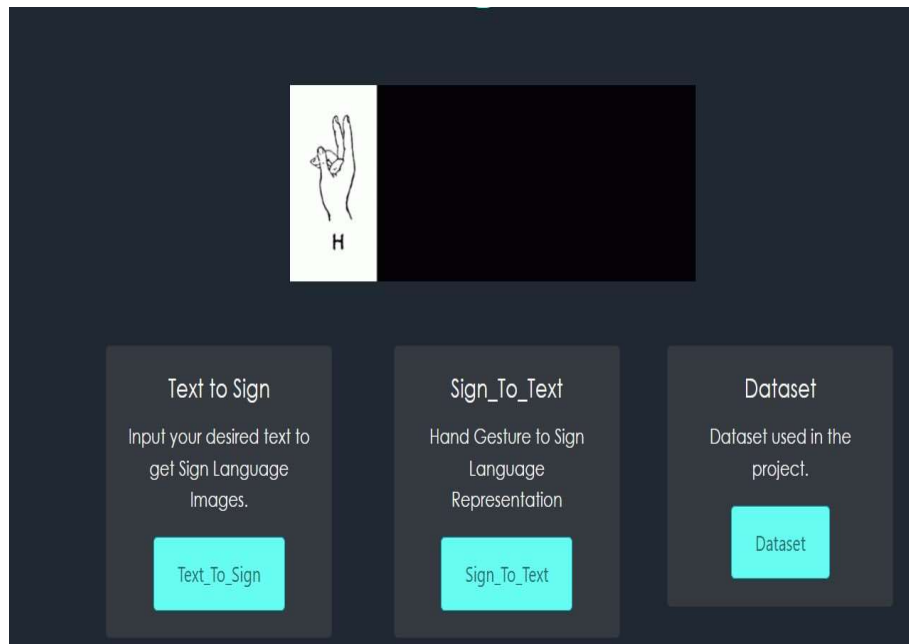
Preparing dataset was rather a lengthy process. The videos clips are challenging, because they capture the complex dynamics of the real world. While the gesture is easy to recognize for humans, it is difficult to understand for a computer because the video footage contains sub-optimal lighting conditions and background noise.

6.3 Training

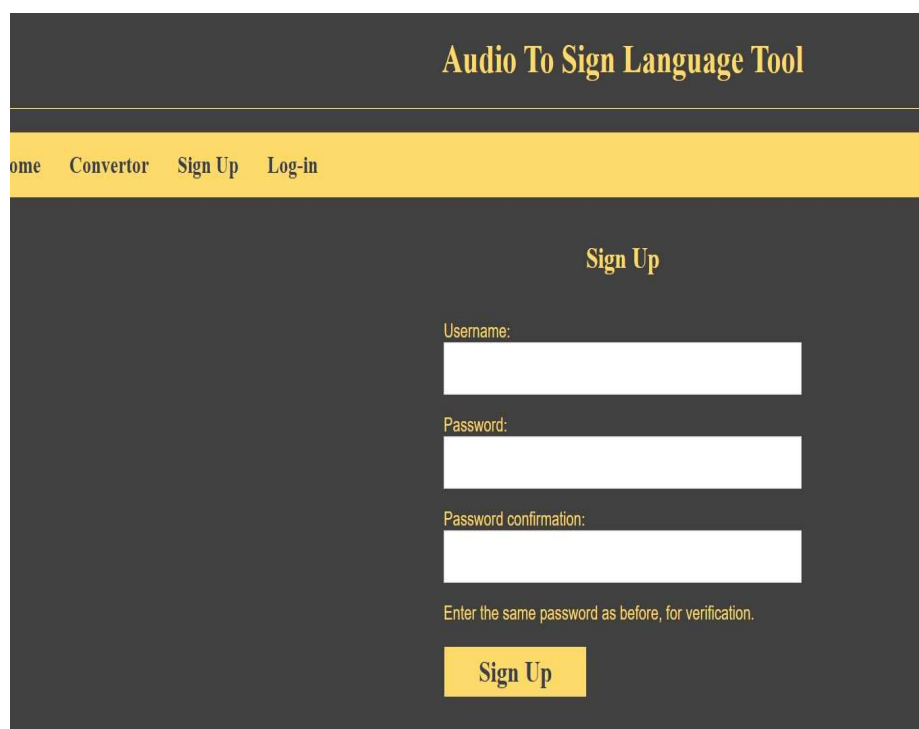
Training such type of complex model is a difficult task altogether. After researching and trying all of the possible information we have come to the conclusion that training in more passes, that is, epochs overfits the model. Also, there should be distinct non-overlapped data elements in the train and test subsets, so as to make the model more robust. In the training data there should be different classes to make model more diverse.

6.4 Algorithm

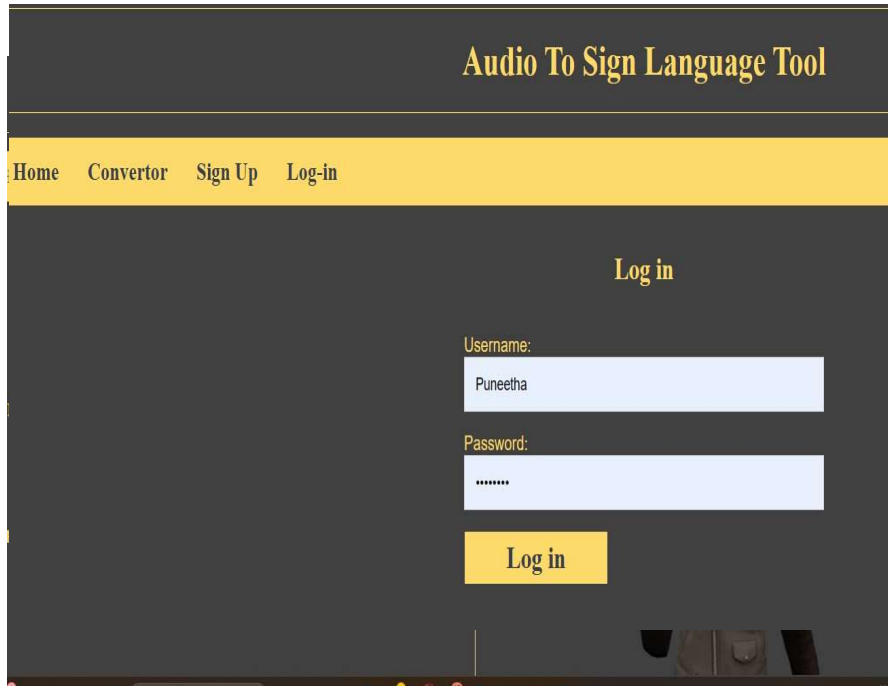
This project is designed to develop a comprehensive device that uses advanced machine-learning knowledge state-of-the-art for sign language text and voice for sign language prediction. This flexible pipeline consists of two foremost components, the primary module focuses on converting hand gestures into textual content, while the second module focuses on translating spoken or written phrases into gestures. The use of present-day techniques, consisting of deep learning techniques in modern fashions which include convolutional neural networks (CNN) and recurrent neural networks (RNN), we search for instantaneous and correct predictions and factors in many languages. By means of leveraging the strength of these algorithms, our intention is to bridge the space in conversation and promote integration and accessibility for folks who use languages and attention to speaking or writing.



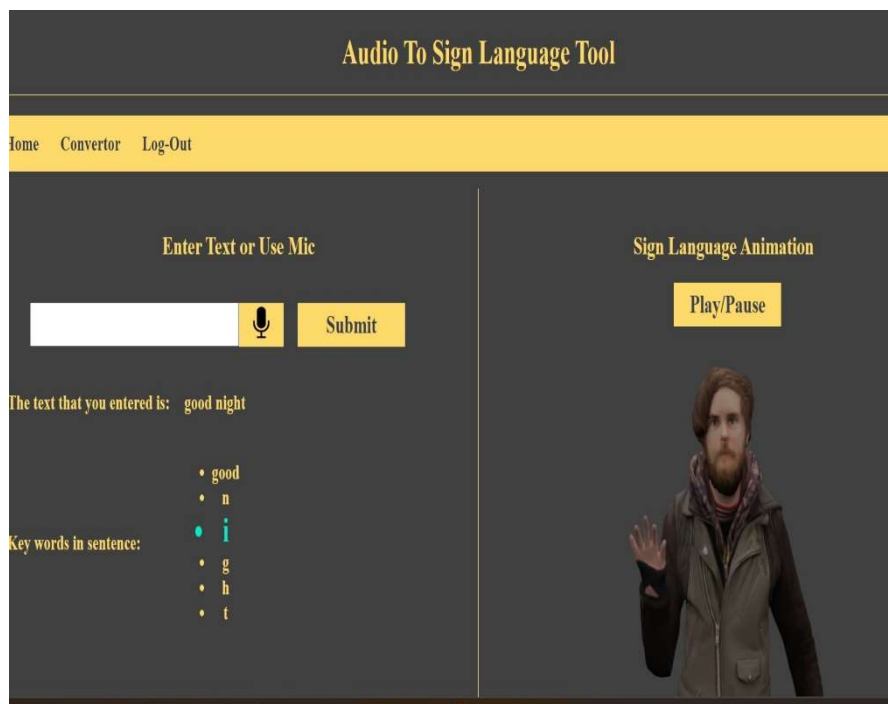
Snapshot 1 Home Page



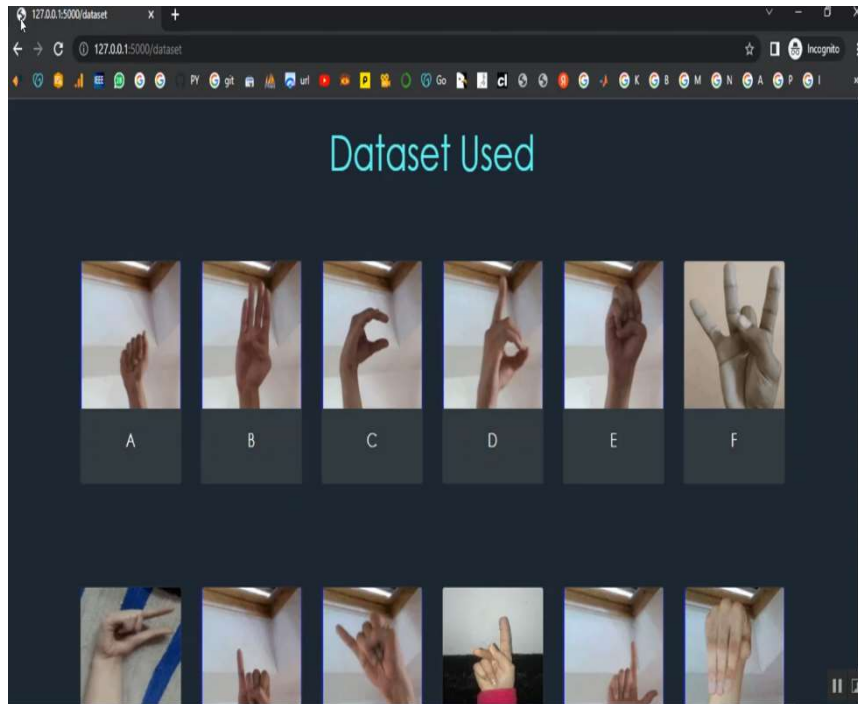
Snapshot 2 Sign up page



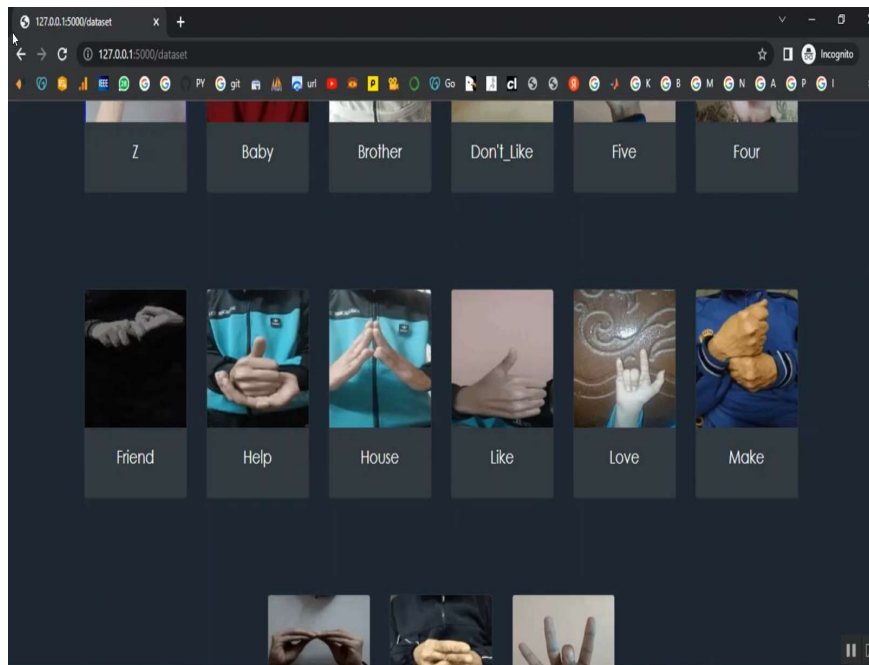
Snapshot 3 Login Page



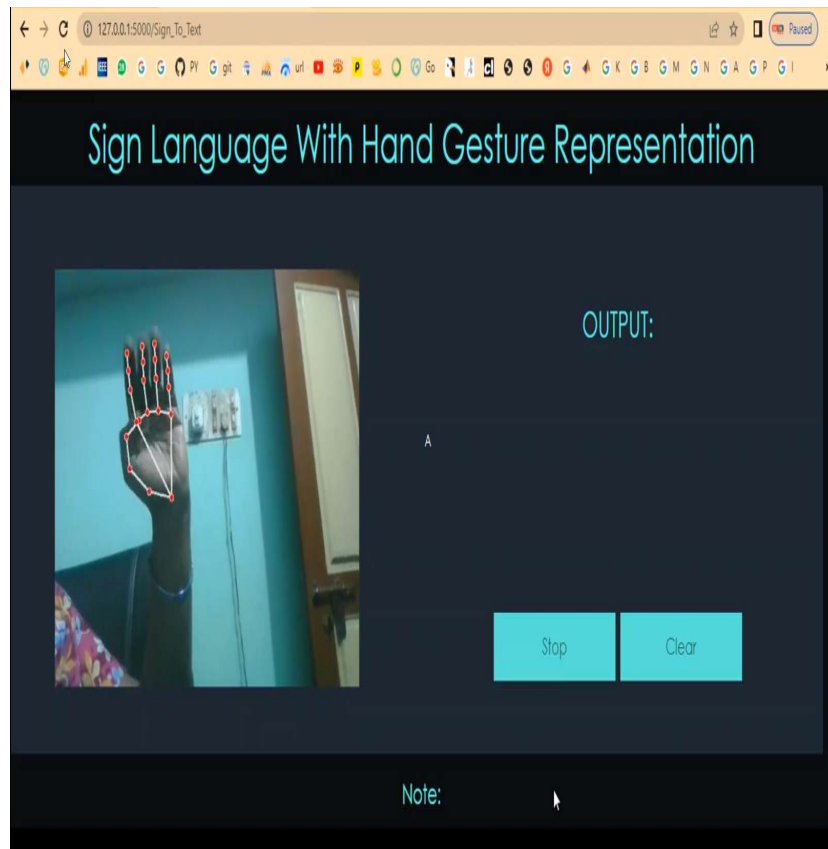
Snapshot 4 Audio to Sign Language Interface



Snapshot 5 Datasets used for alphabets



Snapshot 6 Datasets used for words



Snapshot 7 Sign Language to Text convertor interface

The integration of machine learning into sign language translation holds immense promise for breaking down communication barriers and fostering greater accessibility and inclusivity for individuals with hearing impairments. Through the development of advanced prediction systems capable of translating sign language gestures into text and audio inputs into sign language, we can empower individuals to communicate effectively across diverse linguistic modalities. As machine learning algorithms continue to evolve and improve in accuracy and efficiency, we anticipate the emergence of more sophisticated translation systems that offer real-time, contextually rich communication experiences. These advancements will not only enhance communication accessibility but also promote greater understanding and collaboration in various settings, from educational environments and workplaces to social interactions and public spaces.

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